

(Please write your Roll No. immediately)

Roll No. 25210403619

Mid Term Examination

VI Semester [B. Tech.] March 2020

Paper Code: ETEC-310

Subject: Data Communication &
Networks (Common to MAE, EEE, IT)

Time: 1½ hr

Max. Marks: 30

- Note: 1. Attempt Q1 which is compulsory and any two more questions from the remaining.
2. Necessary data may be assumed and the same may be clearly indicated.

Q1. Answer the following questions briefly.

[5 × 2 = 10 marks]

- What is the difference between network layer delivery and transport layer delivery?
- What is the length of a bit in a channel with a propagation speed of 2×10^8 m/s if the channel bandwidth is 20 Mbps?
- Why does a circuit switched network need end to end addressing during the setup and teardown phases?
- How many number of signal level is needed to transmit 128 kbps over noiseless channel with bandwidth of 20 kHz?
- A system has an n-layer protocol hierarchy. Applications generate messages of length M bytes. At each of the layers an h-byte header is added. What fraction of the network bandwidth is filled with headers?

Q2. a) Discuss the concept of Piggybacking? Consider the use of 1000 bit frames on a 1 Mbps satellite channel. What is the maximum link utilization for? i.) Stop-and-wait ARQ ii.) Continuous ARQ with Window size 7 iii.) Continuous ARQ with Window size 127.

[5 marks]

b) Compare and Contrast between ISO-OSI with TCP/IP reference model.

[5 marks]

Q3. a) Explain the operation of CRC error detection method. By means of an example discuss the i.) generation of error detection bits and ii) how the frame that is received is checked for transmission errors? Use the generator polynomial: $x^5 + x^3 + x^2 + 1$.

[5 marks]

b) Discuss in detail about CSMA/CD multiple access standard. A CSMA/CD bus spans a distance of 1.5 kms, if the data rate is 5 Mbps, what is the minimum frame size?

[5 marks]

Q4. a) Compare IEEE 802.4 and IEEE 802.5 standards based on physical topology, scalability, performance, transmission delay, and frequency.

[5 marks]

b) Write short note on

i) Transmission impairment

ii) PPP frame format

[2 × 2.5 = 5 marks]